

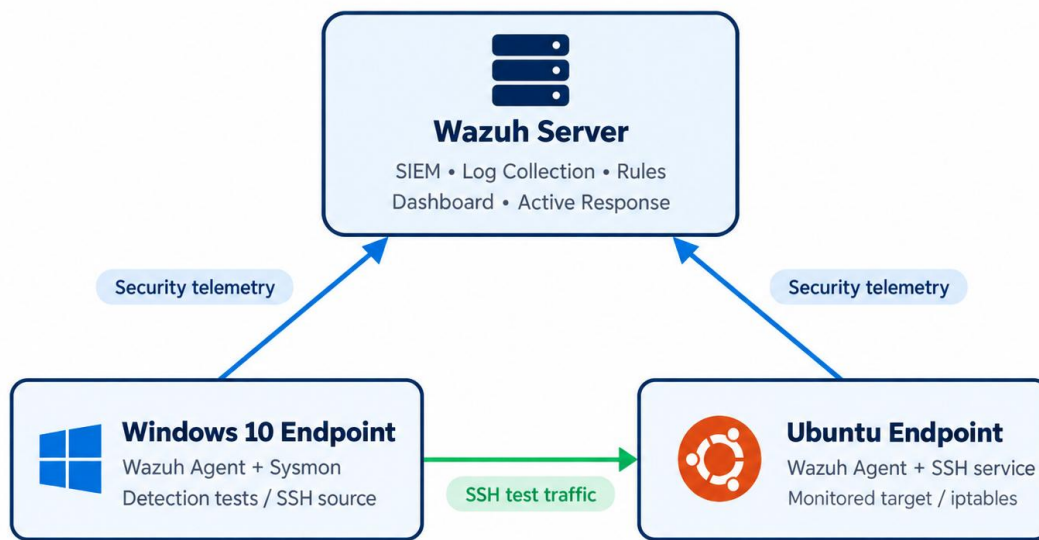
Wazuh SOC Home Lab

Security Monitoring, Detection Engineering & Automated Response

CYBERSECURITY PORTFOLIO PROJECT

Project focus Build a controlled SOC lab, collect endpoint telemetry, create custom detection rules, validate alerts and automate response to SSH brute-force activity.

SOC Home Lab Architecture



Virtualization platform: VMware

High-level lab architecture.

Filipe Dias

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All testing was performed in a controlled VMware lab environment using private network addresses.

1. Project Overview

The purpose of this project was to gain practical, entry-level experience with Security Operations Center (SOC) workflows and to better understand how security monitoring, detection and response operate in practice. The lab was built around Wazuh and focused on connecting endpoints, collecting telemetry, creating detection logic, validating alerts and responding to suspicious activity.

Portfolio value The project demonstrates hands-on exposure to SIEM administration, endpoint telemetry, custom rule development, alert triage, MITRE ATT&CK mapping and active response rather than only theoretical knowledge.

Objectives

- Deploy and operate a small Wazuh-based monitoring environment.
- Connect Windows 10 and Ubuntu endpoints through Wazuh agents.
- Use Sysmon on Windows to enrich endpoint telemetry.
- Create and troubleshoot custom detection rules.
- Validate detections using controlled SSH, account-management and PowerShell tests.
- Build a basic SOC dashboard for recurring security events.
- Configure automated response to repeated failed SSH authentication attempts.

Lab Architecture

SOC Home Lab Architecture

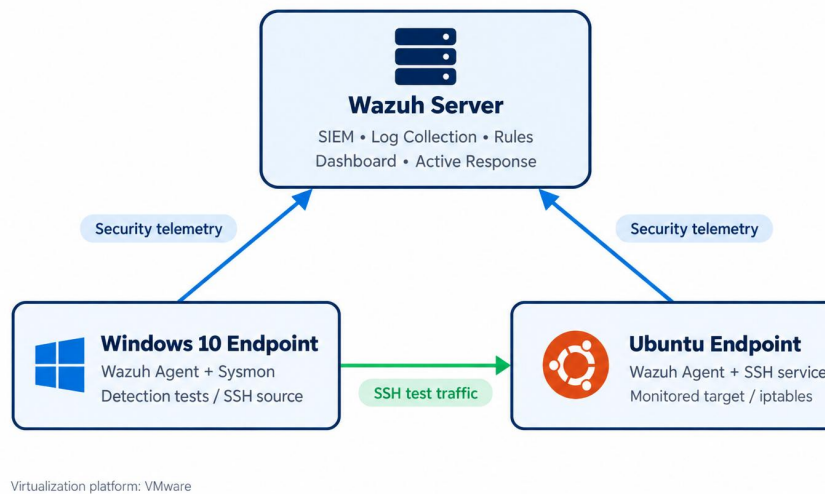


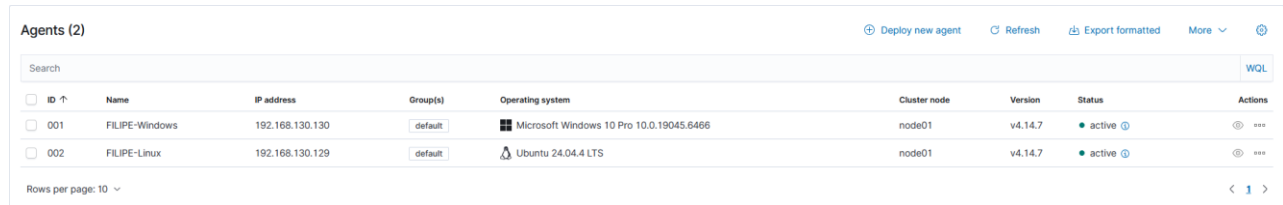
Figure 1. Windows and Ubuntu endpoints forward security telemetry to Wazuh. The Windows endpoint was also used to generate controlled SSH test traffic toward Ubuntu.

Technology Stack

Wazuh	SIEM, log collection, alerting, custom rules and Active Response
Sysmon	Additional Windows endpoint telemetry using Olaf Hartong's community configuration from GitHub, without modification
Windows 10	Monitored endpoint and source of controlled test activity
Ubuntu 24.04.4 LTS	Monitored Linux endpoint and SSH target
VMware	Virtualization platform for the isolated home lab

2. Deployment & Monitoring

Two monitored endpoints were enrolled in Wazuh. The console showed both agents as active, confirming that logs could be collected from the Windows and Ubuntu systems. The Wazuh agent version displayed in the environment was 4.14.7.



The screenshot shows the 'Agents (2)' page in the Wazuh console. It features a search bar, a 'WQL' button, and a table of agents. The table has columns for ID, Name, IP address, Group(s), Operating system, Cluster node, Version, Status, and Actions. Two agents are listed: '001 FILIPE-Windows' and '002 FILIPE-Linux', both with status 'active'.

ID	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
001	FILIPE-Windows	192.168.130.130	default	Microsoft Windows 10 Pro 10.0.19045.6466	node01	v4.14.7	active	
002	FILIPE-Linux	192.168.130.129	default	Ubuntu 24.04.4 LTS	node01	v4.14.7	active	

Figure 2. Wazuh agents: FILIPE-Windows (Windows 10 Pro) and FILIPE-Linux (Ubuntu 24.04.4 LTS), both active.

Windows Telemetry with Sysmon

Sysmon was installed on the Windows endpoint using Olaf Hartong's community configuration downloaded from GitHub. The configuration was used without modification. The goal was to provide additional endpoint telemetry that Wazuh could ingest and correlate with security events. No custom Sysmon tuning was applied in this version of the lab, keeping the implementation aligned with the project objective of building a solid baseline first.

SOC Dashboard

A custom dashboard named “Filipe Basic SOC Activity Overview” was created to surface recurring activity relevant to the lab. The dashboard included failed Windows logons, Linux failed SSH authentication activity and Windows account changes over time.

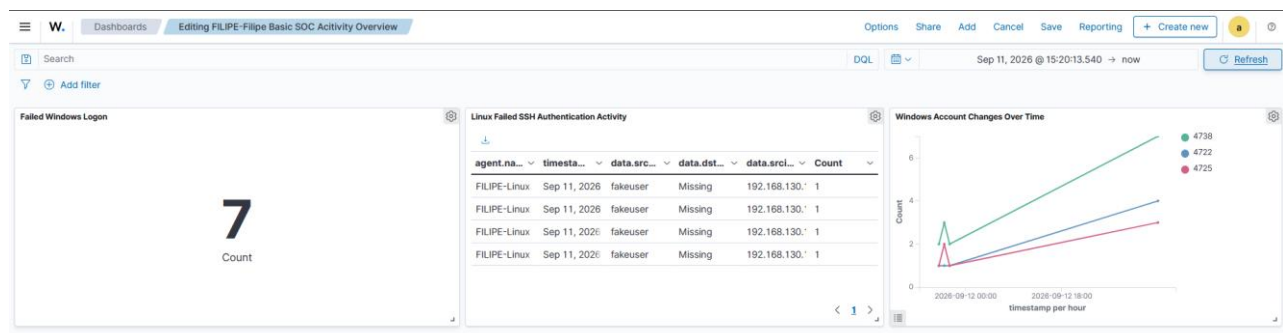
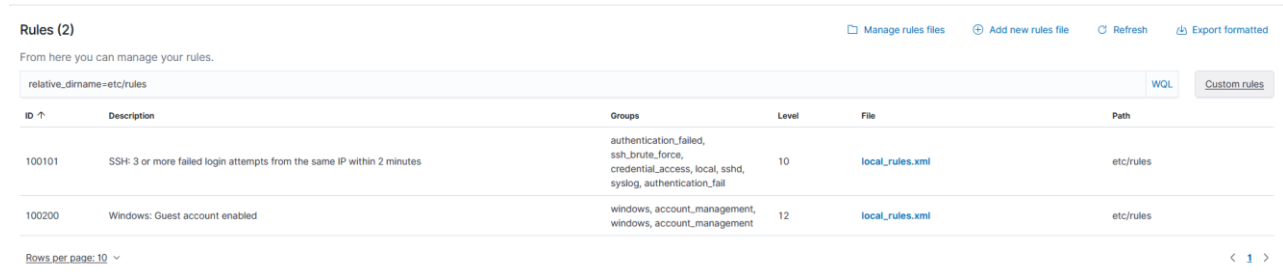


Figure 3. Custom SOC dashboard summarizing authentication failures and Windows account changes.

3. Custom Detection Rules

Custom rules were added to `local_rules.xml` and iteratively tested until the expected events generated the correct alerts. This was one of the main troubleshooting areas of the project and required validating parent rule IDs, event fields, correlation conditions and dashboard output.



The screenshot shows the Wazuh Rules interface. At the top, there are links for 'Manage rules files', 'Add new rules file', 'Refresh', and 'Export formatted'. Below this is a search bar with the text 'relative_dirname=etc/rules' and a 'WQL' button. A 'Custom rules' button is also visible. The main table lists two rules:

ID ↑	Description	Groups	Level	File	Path
100101	SSH: 3 or more failed login attempts from the same IP within 2 minutes	authentication_failed, ssh_brute_force, credential_access, local, sshd, syslog, authentication_fail	10	local_rules.xml	etc/rules
100200	Windows: Guest account enabled	windows, account_management, windows, account_management	12	local_rules.xml	etc/rules

At the bottom, there is a 'Rows per page: 10' dropdown and a pagination control showing '< 1 >'.

Figure 4. Two custom rules visible in the Wazuh rules interface.

Rule ID	Level	Purpose	MITRE / Use
100101	10	3+ failed SSH logins from the same source IP within 120 seconds	T1110 - Brute Force
100200	12	Windows Guest account enabled	T1098 - Account Manipulation

Correlation Logic - SSH Brute Force

```
<rule id="100101" level="10" frequency="3" timeframe="120">
  <if_matched_sid>5760</if_matched_sid>
  <same_srcip />
  <description>SSH: 3 or more failed login attempts from the same IP within 2 minutes</description>
  <mitre><id>T1110</id></mitre>
</rule>
```

The key learning point was moving from a single authentication-failure event to behavioral correlation: the rule only escalates after repeated failures from the same source IP inside a defined time window.

4. Detection Scenario - SSH Brute Force

Scenario

Controlled failed SSH authentication attempts were generated from the Windows endpoint (192.168.130.130) against the Ubuntu endpoint (192.168.130.129) using a non-valid account. The purpose was to validate that repeated failures from the same source would trigger the custom correlation rule.

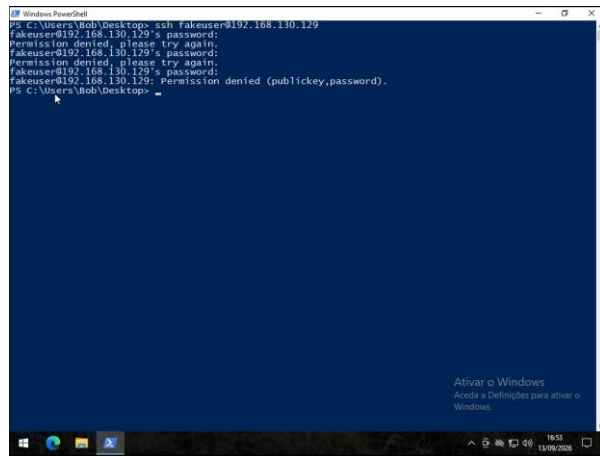


Figure 5. Repeated SSH login failures generated from Windows PowerShell.

Sep 13, 2024 8:15:00 AM UTC - 3 or more failed login attempts from the same IP within 2 minutes

Expanded document

View surrounding documents (7) View single document (7)

Index	wazuh-alerts-4.x-2024-09-13
agent.id	880
agent.ip	192.168.130.130
agent.name	FILIPPE-LINUX
data.destuser	fakeuser
data.srcip	192.168.130.130
data.msgtype	event
decoder.name	sshd
decoder.parent	sshd
full_log	Sep 13 14:04:08 ubuntu sshd[3040]: Failed password for filipe from 192.168.130.130 port 49020 sshd
id	100000000-1000001
timestamp	nan

Figure 6. Wazuh alert generated after the correlation threshold was reached.

Observed Evidence

Source host	FILIPPE-Windows - 192.168.130.130
Target host	FILIPPE-Linux - 192.168.130.129
Service	SSH / sshd
Detection rule	100101 - 3 or more failed login attempts from the same IP within 2 minutes
Severity	Level 10
MITRE ATT&CK	T1110 - Brute Force
Verdict	True Positive - controlled test activity

The Wazuh event exposed the source IP, destination user, monitored agent and raw sshd log entry, making it possible to confirm that the alert represented the activity intentionally generated in the lab.

5. Detection Scenario - Windows Account Manipulation

Scenario

The built-in Windows Guest account (“Convidado” in the Portuguese Windows installation) was enabled on the monitored Windows 10 endpoint. A custom Wazuh rule was designed to detect this account-management event and raise a high-severity alert.

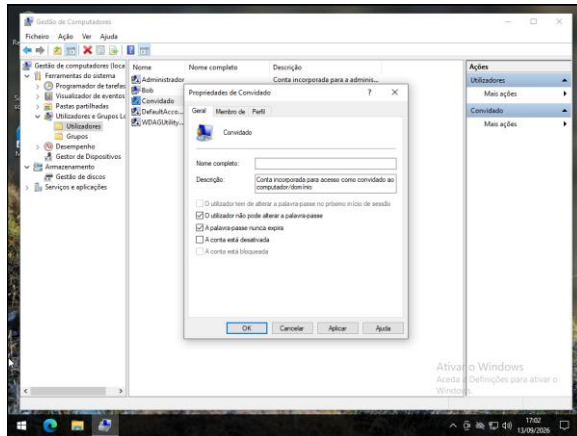


Figure 7. Windows Computer Management showing the Guest account enabled.



Figure 8. Wazuh alert: “Windows: Guest account enabled”.

Detection Logic

```
<rule id="100200" level="12">
  <if_sid>60110</if_sid>
  <field name="win.eventdata.targetUserName">Convidado</field>
  <description>Windows: Guest account enabled</description>
  <mitre><id>T1098</id></mitre>
</rule>
```

Monitored endpoint	FILIPE-Windows - 192.168.130.130
Custom rule	100200
Severity	Level 12
MITRE ATT&CK	T1098 - Account Manipulation
Outcome	The event matched the localized account name “Convidado” and generated the expected alert.

Troubleshooting lesson The Windows installation was in Portuguese, so the event field contained “Convidado” rather than “Guest”. Correcting the expected targetUserName was necessary for the rule to trigger reliably.

6. Detection Scenario - PowerShell Activity Monitoring

PowerShell activity was generated on the Windows endpoint using the harmless `whoami` command. This scenario was used to verify that Sysmon and Wazuh were collecting useful PowerShell-related telemetry. Because the activity was benign and controlled, this scenario is documented as monitoring rather than as a “PowerShell attack”.

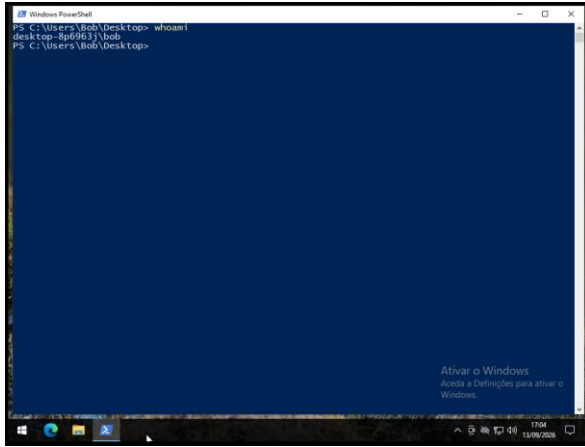


Figure 9. Benign PowerShell activity generated with the `whoami` command.

Aug 15, 2024 9:17:00:25:000 Executable file dropped in folder commonly used by malware

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Table .JSON

._index	wazuh-alerts-4.x-2024.08.15
_agent_id	001
_agent_ip	192.168.124.100
_agent_name	10.100.0.100
data.event.data.timestamp	2024-08-15 16:00:25.000
data.event.data.image	C:\Windows\System32\WindowsPowerShell\v1.0\PowerShell.exe
data.event.data.processid	(4432x00-c94b-ea4b-7940-000000000000)
data.event.data.processid	8440
data.event.data.ridename	technique_t1059.001, technique_namepowershell
data.event.data.targetfilename	C:\Users\Bob\AppData\Local\Temp\1\p08r9p9h3zyTest_wesp-081_rdb.pdf
data.event.data.user	DESKTOP-8P6963J\Bob

Figure 10. Wazuh/Sysmon telemetry showing PowerShell-related activity.

Endpoint	FILIPE-Windows
Process observed	PowerShell.exe
User context	DESKTOP-8P6963J\Bob
MITRE mapping in telemetry	T1059.001 - PowerShell
Purpose	Validate visibility into PowerShell execution and related endpoint events

The captured Wazuh event showed PowerShell as the originating process and included a PowerShell-related MITRE technique mapping. The evidence confirms visibility into command-line activity, although the specific alert shown should be interpreted as supporting telemetry rather than proof that the benign `whoami` command itself was malicious.

7. Automated Response to SSH Brute Force

After validating the SSH brute-force detection, Wazuh Active Response was configured so that rule 100101 would trigger the built-in `firewall-drop` command on the Ubuntu endpoint. This extended the project from detection-only monitoring to automatic containment.

```
<active-response>
  <disabled>no</disabled>
  <command>firewall-drop</command>
  <location>local</location>
  <rules_id>100101</rules_id>
</active-response>
```

Figure 11. Active Response configuration linking firewall-drop to rule 100101.

Sep 13, 2026 @ 17:14:26.791 Host blocked by firewall-drop Active Response

Expanded document

Table JSON

._index	wazuh-alerts-4.x-2026.09.13
_agent.id	002
_agent.ip	192.168.130.129
_agent.name	FILIPPE-Linux
_data.command	add
_data.dstuser	filipe
_data.origin.module	wazuh-execd
_data.origin.name	node01
_data.parameters.alert_agent.id	002
_data.parameters.alert_agent.ip	192.168.130.129
_data.parameters.alert_agent.name	FILIPPE-Linux

Figure 12. Wazuh confirms that the host was blocked by firewall-drop Active Response.

```
filipe@ubuntu:~$ sudo iptables -L -n --line-numbers
[sudo] password for filipe:
Chain INPUT (policy ACCEPT)
num target prot opt source destination
1 DROP 0 -- 192.168.130.130 0.0.0.0/0

Chain FORWARD (policy ACCEPT)
num target prot opt source destination
1 DROP 0 -- 192.168.130.130 0.0.0.0/0

Chain OUTPUT (policy ACCEPT)
num target prot opt source destination
filipe@ubuntu:~$
```

Figure 13. Ubuntu iptables shows DROP rules for the source IP.

Response Flow

1. Detection	Three or more failed SSH logins from the same source in 120 seconds.
2. Rule trigger	Custom rule 100101 reached severity level 10.
3. Active Response	Wazuh executed firewall-drop locally on the monitored Ubuntu host.
4. Containment	The source IP 192.168.130.130 was added to DROP rules in iptables.
5. Validation	Both Wazuh event data and the Linux firewall state confirmed the response.

Key result The lab demonstrated an end-to-end SOC workflow: event generation → detection → correlation → alert → automated containment → validation.

8. Incident Investigation & Triage

The SSH brute-force scenario was used as the primary investigation case because it contained a clear source, target, sequence of events, custom correlation logic and a response action.

Investigation Summary

Alert	SSH: 3 or more failed login attempts from the same IP within 2 minutes
Source IP	192.168.130.130
Affected host	FILIFE-Linux - 192.168.130.129
Observed behavior	Repeated failed SSH authentication attempts from one source
Validation	Raw sshd logs and Wazuh event fields matched the controlled test
Classification	True Positive
Response	Source IP blocked using Wazuh Active Response / firewall-drop

Challenges & Troubleshooting

- Creating and applying custom Wazuh rules required repeated testing and correction before alerts triggered as expected.
- The localized Windows account name had to be matched correctly (“Convidado” instead of “Guest”).
- Correlation logic for SSH required validating the correct parent rule and ensuring events were grouped by the same source IP.
- Dashboard panels required adjustment until the desired event types and fields were represented clearly.
- Active Response was validated independently by checking the resulting iptables DROP rules on Ubuntu.

Operational mindset the most useful part of the project was not simply installing tools, but repeatedly testing assumptions against logs and adjusting configuration until the observed behavior matched the intended detection logic.

9. Skills Demonstrated & Lessons Learned

Skills Demonstrated

- Wazuh deployment and endpoint agent management
- Windows and Linux security monitoring
- Sysmon telemetry collection
- Custom Wazuh rule development and troubleshooting
- Event correlation and severity assignment
- MITRE ATT&CK mapping (T1110, T1098, T1059.001)
- SOC dashboard creation
- SSH authentication analysis
- True-positive validation and basic incident triage
- Automated containment using Wazuh Active Response and Linux iptables

Lessons Learned

This project provided practical experience configuring Wazuh and its agents, building custom detection rules and implementing automated blocking for repeated SSH authentication failures. Troubleshooting the rules and dashboards was particularly valuable because it required understanding the event data rather than simply following installation steps.

The experience also increased my interest in SOC and Blue Team work. At the same time, I want to continue exploring other cybersecurity areas before deciding which specialization best fits my strengths and long-term goals.

Current Limitations & Next Steps

- The lab currently uses Olaf Hartong's community Sysmon configuration without modification; future work could introduce custom tuning and compare noise versus detection value.
- Expand detection coverage with additional Windows authentication, process and persistence scenarios.
- Create more structured incident reports with timelines and indicators of compromise.
- Add network telemetry or IDS data to correlate endpoint events with network activity.
- Continue refining dashboards to support faster triage and higher-value SOC metrics.

Conclusion

The Wazuh SOC Home Lab successfully demonstrated a complete baseline security-monitoring workflow: endpoint onboarding, telemetry collection, dashboarding, custom detection engineering, controlled validation, incident triage and automated response. The project provides practical evidence of foundational SOC/Blue Team skills and establishes a platform that can be expanded with more advanced detection and investigation scenarios.

Portfolio takeaway A recruiter or interviewer can use this project to see not only which tools were used, but how alerts were designed, tested, investigated and acted upon in a controlled environment.